AIP | Review of Scientific Instruments

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Citation: Review of Scientific Instruments **48**, 359 (1977); doi: 10.1063/1.1135027
View online: http://dx.doi.org/10.1063/1.1135027
View Table of Contents: http://scitation.aip.org/content/aip/journal/rsi/48/3?ver=pdfcov
Published by the AIP Publishing

Qualitative agreement with the preceding equations was demonstrated by detecting outgassing of the Pyrex bell jar. The top of the bell jar was flame heated to increase the local outgassing rate of a spot inside the bell jar about 53 cm above the leak telescope. This heating raised the pressure in the bell jar from 1.5×10^{-7} to 2.5×10^{-7} Torr. Assuming that the angular distribution of the outgoing molecules follows the cosine law, and since the pumping speed of the bell jar is known ($S \approx 200$ l/sec), an absolute estimate of the outgassing flux can be made. The measured response $(|\Delta n_2|/n_2) = 3 \times 10^{-3}$ corresponds to outgassing from an area of a circle 2 cm in diameter, a very reasonable result.

The sensitivity demonstrated in this experiment is adequate for our purposes. The signal-to-noise ratio in the pump current was about 2:1 ($\omega/2\pi = 9.5$ Hz and lock-in time constant RC = 1 sec). The noise was dominated by appendage pump phenomena. Scanning the noise power spectrum with a tuned amplifier showed that the noise power per unit bandwidth was flat between about 2 and 30 Hz, above which it fell roughly as the inverse frequency squared. At various pressures small relaxation oscillations occur and those pressure regions were avoided in taking data.

The ideal density sensor would be a particle counter of small volume for which shot noise dominates. Consider an ion gauge with an average current density of 10 mA/cm² as a quantitative illustration. The mean time τ_I for ionization of a neutral drifting through this electron beam will be of the order of 0.1 sec. The rms signal-to-noise ratio would be

$$1/2\left(\frac{n_2}{\tau_I \Delta f V_2}\right)^{1/2}\left(\frac{\Delta\Omega S_1}{2\pi\omega}\right)f(\theta),$$

where Δf is the bandwidth of the ion current detector. In our system, for $\Delta f = 1$ Hz, the leak dN/dt corresponding to unity signal-to-noise would be less than 10^{-11} Torr l/sec, comparable with commercial helium leak detectors.

Scanning a large vacuum chamber for leaks may be a lengthy process and special strategies for finding large leaks before high resolution searches for lesser leaks may be desirable. Analog or digital instrumentation for displaying the scan data on an $x-y$ recorder or CRT display would make it possible to present a "vacuum image" of the vacuum system surfaces in which all gas sources are identified. If a mass spectrometer head were used as the detector, gas identification could be added to the vacuum image (color TV).

[1] A preliminary description of this device is reported in J. C. Wesley, E. S. Ensberg, and T. H. Jensen, Bull. Am. Phys. Soc. **21**, 1088 (1976). An earlier example of leak detection through molecular flow analysis by means of a system of ionization gauges and gas analyzers is discussed by H. K. F. Ehlers, J. Spacecr. Rockets **7**, 480 (1970); and *Space Simulation* (proceedings of a NASA Symposium), Publication NASA SP-298 (Washington, DC, 1972), Paper No. 70.

FET Q switch for pulsed NMR

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(Received 20 September 1976; in final form, 2 December 1976)

A Q switch which shortens the ringing time in pulsed NMR equipment is described. It uses a field-effect transistor as the Q-switching element. A convenient method of generating the control signal is shown. The device performs well and is simple to construct and adjust.

In pulsed NMR work the nuclear induction signal typically is obscured for many microseconds after the transmitter pulse by the ringing of the input tuned circuit. Because the sensitivity of the apparatus is improved by using a high-Q input tuned circuit, the most satisfactory solution to the ringing problem is to employ a Q switch on the input circuit. For a short time following the transmitter pulse the Q of the input circuit is lowered, causing it to ring down quickly. Then the Q is increased to observe the nuclear signal.

Circuits using diodes as Q switches for this purpose have been described;[1-4] in these circuits the impedance of the diode is varied by controlling the bias current through the diode. A major problem with using diodes for this purpose is that they are two-lead devices: whenever the Q is changed, the change in diode bias current flows through the input tuned circuit and causes more ringing. Spokas's circuit[2] is a balanced arrangement which prevents this; it works well but requires careful balancing for proper operation.

The field-effect transistor (FET) is a nearly ideal device for use as a Q switch: the resistance of the channel is controlled by the gate voltage. Because the gate is isolated from the channel (unlike bipolar transistors) and the FET channel can operate with no bias (unlike vacuum tubes), the Q of the tuned circuit can be controlled without the generation of currents that would prolong ringing.

The scheme reported here employs a FET as a Q switch. It is effective in reducing ringing and is simpler than previous designs. It requires no balancing and no careful shaping of the control waveform. A convenient method of generating the control voltage is shown.

Figure 1 shows a single-coil NMR tuning circuit incorporating the FET Q switch; the values shown are for 20 MHz operation. Diodes D_3 and D_4 decouple the transmitter from the input tuned circuit; D_5 and D_6 protect Q_1 and the preamplifier input stage. L is the NMR sample coil; L and C together form the input tuned circuit. During transmitter pulses D_1 is shunted across L and C through diodes D_5 and D_6; L_1 is chosen to resonate this additional capacitance. R_1 is the bias return for the preamplifier. Q_1 is the Q-switching (n-channel) FET and Q_2 is the control transistor. L_2 is a one-turn link loosely coupled to L_1. Diodes D_1 and D_2 protect Q_2.

When the transmitter is pulsed on, Q_2 is turned on during negative half-cycles of Q_2's base voltage. The collector voltage of Q_2 quickly rises to +9 V, where it remains throughout the transmitter pulse duration. When the transmitter pulse ends, Q_2 stops conducting. The collector voltage of Q_2 and the gate voltage of Q_1 decrease to −9 V at a rate determined by the 25 kΩ variable resistor, the 820 Ω resistor, and the 500 pF capacitor. As long as the gate voltage of Q_1 is more positive than the pinchoff voltage of Q_1 (V_p, about −3 V), the source-drain path appears as a low resistance (about 200 Ω) in series with C_1, together shunting the input tuned circuit. This decreases the Q of the input circuit. When the gate voltage is more negative than V_P, Q_1 is turned off and the Q of the input circuit increases. Of course, the Q does not switch abruptly but changes gradually as the gate voltage passes through V_P.

To use the Q switch with a double-coil NMR apparatus, omit diodes D_3 and D_4 and short across the parallel combination of C_1 and R_1.

Many radio-frequency, n-channel FETs worked well in the circuit of Fig. 1. We have used the 3N152 MOSFET (by RCA) and the following junction FETS: MPF102 (Motorola), HEP F0015 (Motorola), and ECG132 (Sylvania). The devices should be selected for low zero-bias resistance from source to drain (R_{ON}). It is easy to find units with R_{ON} values between 200 and 300 Ω. When using junction FETs in this circuit, a 100 kΩ resistor should be placed in series with the gate of the junction FET to limit the gate current drawn by the device.

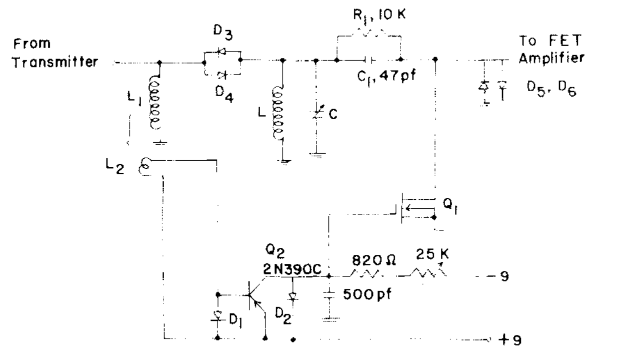

FIG. 1. Circuit of the Q switch, used with a single-coil NMR probe at 20 MHz. All diode are 1N914B, 1N4454A, or 1N4864.

To obtain the lowest Q in the low-Q state (FET biased on), C_1 should be chosen so that $(\omega C_1)^{-1}$ is roughly equal to or less than R_{ON} of Q_1 ($\omega \equiv 2\pi \times$ resonance frequency). This is because C_1 and Q_1 together in series load the input tuned circuit. If $(\omega C_1)^{-1} = 200\ \Omega$, a transmitter voltage of 1000 V will generate a 5 A current through C_1, D_5, and D_6. This current can generate voltages large enough to destroy Q_1 and/or the preamplifier if there is a sizable lead inductance associated with D_5 and D_6. The preamplifier connection, D_5, D_6, and Q_1, should be wired with short leads and should be located within a few centimeters of each other.

The circuit can be run from two 9 V batteries. There is no current drawn from the batteries except during transmitter pulses, so they should have a long life. Also, the circuit can be run from just one battery with little change in performance by grounding the point marked +9 in Fig. 1.

Adjustment of the circuit is simple. After tuning in normal fashion R is varied to minimize the receiver dead time. If R is too small, the input circuit regains its high Q before the ringing is completed. If R is too large the Q remains low after the ringing has completed; the nuclear signal will not be detected until the Q returns to its higher value.

Two versions of the FET Q switch have been built, one operating at 7.55 MHz and the other at 20 MHz.

The 7.55 MHz version was used with a double-coil NMR probe. The receiving coil had a Q of 50 when the FET was biased off. Without the Q switch, the coil ringing persisted (before being obscured by receiver noise) for 20 μsec after the end of a 4 μsec transmitter pulse. With the Q switch the ringing lasted for 9 μsec.

The 20 MHz version was used with a single-coil NMR probe. The transmitter $\pi/2$ pulses were 5 μsec long; the coil Q was 63 with the Q switching FET biased off. When the Q switch was employed, the ringing time decreased from 10 to 5 μsec. We have used the Q switch at 20 MHz in conjunction with signal averaging (up to 8000 averages); the increased sensitivity due to averaging has not revealed any longer-time recovery effects.

The author thanks David Cornell and Nouha Salibi for constructing and testing the 7.55 MHz Q switch. He also thanks R. E. Norberg for reading the manuscript; he gratefully acknowledges the NSF for its support.

* National Science Foundation Graduate Fellow; work supported by a research grant from the National Science Foundation.

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[2] John J. Spokas, Rev. Sci. Instrum. **36,** 1436 (1965).

[3] Irving J. Lowe and Dennis E. Barnaal, Rev. Sci. Instrum. **34,** 143 (1963).

[4] Kenneth E. Kisman and Robin L. Armstrong, Rev. Sci. Instrum. **45,** 1159 (1974).

One-point calibration of Allen–Bradley resistor thermometers, 2–20 K

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(Received 4 October 1976; in final form, 22 November 1976)

Calibration studies of 25 Allen–Bradley BB resistors (220 Ω, 1/8 W) between 2 and 20 K according to $\log R = A + BT^{-P}$ indicate that A, B, and P vary linearly with the resistance at 4.2 K.

The ubiquitous carbon radio resistor is found in every low-temperature laboratory, and the literature is replete with studies of these low-cost, disposable thermometers.[1] The analytical handling of $R-T$ data has been the subject of many papers dealing with empirical formulas,[2] interpolation schemes,[3] and comparisons of various resistor types.[4] The purpose of this note is to report calibration studies on several Allen–Bradley resistors of the same type (220 Ω, ⅛ W BB) in the same temperature range (2–20 K) according to the same empirical formula, viz.,

$$\log R = A + BT^{-P}, \qquad (1)$$

which was first suggested by Clement and Quinnell.[2]

Twenty-five resistors were calibrated in the course of calorimetric measurements.[5] The phenolic resin was ground off the resistors and the leads trimmed to about 1 mm. Four-terminal dc potentiometric measurements were made on both the resistors and a calibrated germanium thermometer.[6] About 15 calibration points were measured on each resistor (full experimental details are given in Ref. 5), and the $R-T$ data were fitted to Eq. (1) using multiple regression analysis methods on a computer. In the range 2–20 K, Eq. (1) fits the $R-T$ characteristic curve with residuals $\Delta T/T \lesssim \pm 0.5\%$ and correlation coefficients >0.999.

No special precautions were taken in preparing these resistors (e.g., tempering in liquid nitrogen or selecting on R_{300}), and the units were obtained from different suppliers. Yet it was noticed that not only did the coefficients show small variations ($2.42 < A < 2.52, 3.0 < B < 3.6, 0.72 < P < 0.81$), but also these coefficients scaled with $R_{4.2}$. Data for the A, B, and P coefficients determined on the 25 different resistors are shown in Fig. 1. The straight lines in Fig. 1 are least-squares fits to linear equations, viz.,

$$\begin{aligned} A &= 2.2148 + 7.0173 \times 10^{-5} R_{4.22} \\ B &= 1.7504 + 4.1888 \times 10^{-4} R_{4.22} \\ P &= 0.5533 + 5.7999 \times 10^{-5} R_{4.22}. \end{aligned} \qquad (2)$$

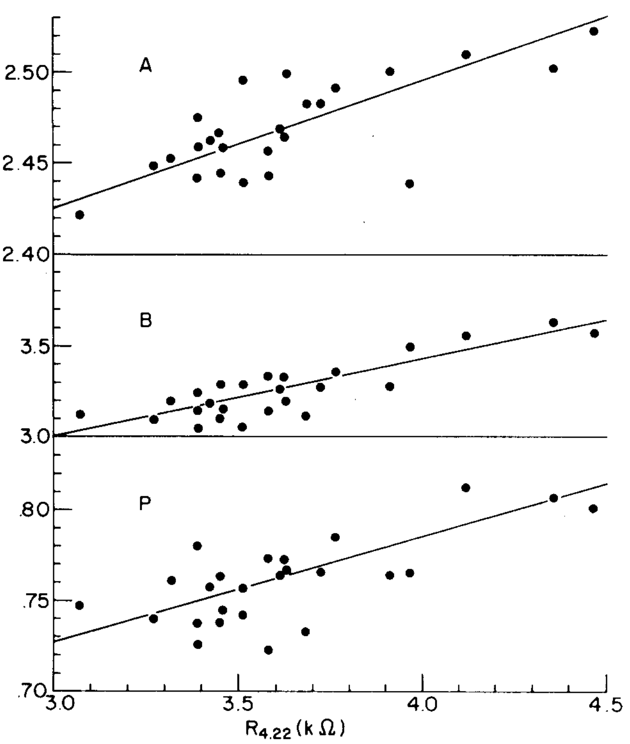

FIG. 1. Variations of the A, B, and P coefficients of Eq. (1) with resistance at 4.2 K for 220 Ω, ⅛ W Allen–Bradley BB resistors calibrated in the range 2–20 K. The straight lines are least-squares linear fits to the data.